

ENEL445 — Gradient Methods Comparison

$$\min f(X) = x_1^2 + 5x_2^2, \quad X_0 = (3, 1)^T, \quad X \in \mathbb{R}^2$$

Shared Quantities at X_0

$$\nabla f(X_0) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix}_{X_0} = \begin{bmatrix} 6 \\ 10 \end{bmatrix} \quad H_f = \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix} \quad H_f^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix}$$

Every method iterates $X_k = X_{k-1} + \alpha_{k-1} p_{k-1}$. Methods differ only in how p_{k-1} is chosen.

Method 1 — Steepest Descent

Direction:

$$p_0 = -\nabla f(X_0) = \begin{bmatrix} -6 \\ -10 \end{bmatrix}$$

Candidate point: $\hat{X}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \alpha \begin{bmatrix} -6 \\ -10 \end{bmatrix} = \begin{bmatrix} 3 - 6\alpha \\ 1 - 10\alpha \end{bmatrix}$

Line search: $\phi(\alpha) = (3 - 6\alpha)^2 + 5(1 - 10\alpha)^2 = 14 - 136\alpha + 536\alpha^2$

$$\frac{d\phi}{d\alpha} = -136 + 1072\alpha = 0 \quad \Rightarrow \quad \alpha_0 = \frac{136}{1072} \approx 0.1269$$

Result: $X_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + 0.1269 \begin{bmatrix} -6 \\ -10 \end{bmatrix} = \begin{bmatrix} 2.239 \\ -0.269 \end{bmatrix}$

Method 2 — Conjugate Gradient (Fletcher–Reeves)

Iteration 1 ($\beta_0 = 0$, no previous direction — identical to steepest descent):

$$p_0 = -\nabla f(X_0) = \begin{bmatrix} -6 \\ -10 \end{bmatrix} \quad \text{Line search gives } \alpha_0 = \frac{136}{1072} \approx 0.1269$$

Result: $X_1 = \begin{bmatrix} 2.239 \\ -0.269 \end{bmatrix}$

Iteration 2 (conjugate correction applied):

Gradient at X_1 : $\nabla f(X_1) = \begin{bmatrix} 2 \times 2.239 \\ 10 \times (-0.269) \end{bmatrix} = \begin{bmatrix} 4.478 \\ -2.687 \end{bmatrix}$

Fletcher–Reeves β_1 :

$$\beta_1 = \frac{\nabla f(X_1)^T \nabla f(X_1)}{\nabla f(X_0)^T \nabla f(X_0)} = \frac{4.478^2 + 2.687^2}{6^2 + 10^2} = \frac{27.27}{136} = 0.2005$$

Conjugate direction:

$$\boxed{p_1 = -\nabla f(X_1) + \beta_1 p_0} = \begin{bmatrix} -4.478 \\ 2.687 \end{bmatrix} + 0.2005 \begin{bmatrix} -6 \\ -10 \end{bmatrix} = \begin{bmatrix} -5.681 \\ 0.682 \end{bmatrix}$$

Candidate point: $\hat{X}_2 = \begin{bmatrix} 2.239 - 5.681\alpha \\ -0.269 + 0.682\alpha \end{bmatrix}$

Line search: $\phi(\alpha) = 5.373 - 27.27\alpha + 34.59\alpha^2$

$$\frac{d\phi}{d\alpha} = -27.27 + 69.18\alpha = 0 \Rightarrow \alpha_1 = \frac{27.27}{69.18} = 0.3942$$

Result: $X_2 = \begin{bmatrix} 2.239 \\ -0.269 \end{bmatrix} + 0.3942 \begin{bmatrix} -5.681 \\ 0.682 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ **exact minimum in 2 iterations** (= n, guaranteed for quadratic f)

Method 3 — Newton's Method**Iteration 1:**

Hessian and inverse at X_0 : $H_f = \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix}$ (2x2 inverse: swap diagonal, negate off-diagonal, divide by det = 20)

$$H_f^{-1} = \frac{1}{20} \begin{bmatrix} 10 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix}$$

Direction:

$$\boxed{p_0 = -H_f(X_0)^{-1} \nabla f(X_0)} = - \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix} \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$$

Candidate point: $\hat{X}_1 = X_0 + \alpha p_0 = \begin{bmatrix} 3 - 3\alpha \\ 1 - \alpha \end{bmatrix}$

Line search: $\phi(\alpha) = (3 - 3\alpha)^2 + 5(1 - \alpha)^2 = 14 - 28\alpha + 14\alpha^2$

$$\frac{d\phi}{d\alpha} = -28 + 28\alpha = 0 \Rightarrow \alpha_0 = 1$$

(Newton's method always gives $\alpha = 1$ for a quadratic objective — no line search needed)

Result: $X_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + 1 \cdot \begin{bmatrix} -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ **exact minimum in one step** (guaranteed for quadratic f)

Method 4 — BFGS (Quasi-Newton)**Iteration 1** ($\tilde{V}_0 = I$, initialised to identity — identical to steepest descent):

$$\boxed{p_0 = -\tilde{V}_0 \nabla f(X_0)} = -I \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -6 \\ -10 \end{bmatrix} \quad \text{Line search: } \alpha_0 \approx 0.1269$$

Result: $X_1 = (2.239, -0.269)^T$

BFGS rank-2 update (build better $\tilde{V}_1$ from curvature information):

Displacement and gradient change:

$$s_0 = X_1 - X_0 = \begin{bmatrix} -0.761 \\ -1.269 \end{bmatrix} \quad y_0 = \nabla f(X_1) - \nabla f(X_0) = \begin{bmatrix} 4.478 \\ -2.687 \end{bmatrix} - \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -1.522 \\ -12.687 \end{bmatrix}$$

Curvature scalar: $\sigma_0 = \frac{1}{y_0^T s_0} = \frac{1}{(-1.522)(-0.761) + (-12.687)(-1.269)} = \frac{1}{17.254} \approx 0.05796$

Rank-2 update formula:

$$\tilde{V}_1 = (I - \sigma_0 s_0 y_0^T) \tilde{V}_0 (I - \sigma_0 y_0 s_0^T) + \sigma_0 s_0 s_0^T$$

With $\tilde{V}_0 = I$, evaluating term by term:

$$\tilde{V}_1 \approx \begin{bmatrix} 1.217 & -0.086 \\ -0.086 & 0.110 \end{bmatrix}$$

(satisfies secant condition $\tilde{V}_1 y_0 = s_0$; converges to H_f^{-1} as iterations proceed)

Iteration 2:

Direction:

$$p_1 = -\tilde{V}_1 \nabla f(X_1) = - \begin{bmatrix} 1.217 & -0.086 \\ -0.086 & 0.110 \end{bmatrix} \begin{bmatrix} 4.478 \\ -2.687 \end{bmatrix} = \begin{bmatrix} -5.681 \\ 0.682 \end{bmatrix}$$

(same direction as CG — for a 2D quadratic, BFGS and CG are equivalent conjugate direction methods)

Candidate point: $\hat{X}_2 = \begin{bmatrix} 2.239 - 5.681\alpha \\ -0.269 + 0.682\alpha \end{bmatrix}$

Line search: $\phi(\alpha) = 5.375 - 27.285\alpha + 34.602\alpha^2$

$$\frac{d\phi}{d\alpha} = -27.285 + 69.204\alpha = 0 \quad \Rightarrow \quad \alpha_1 \approx 0.3942$$

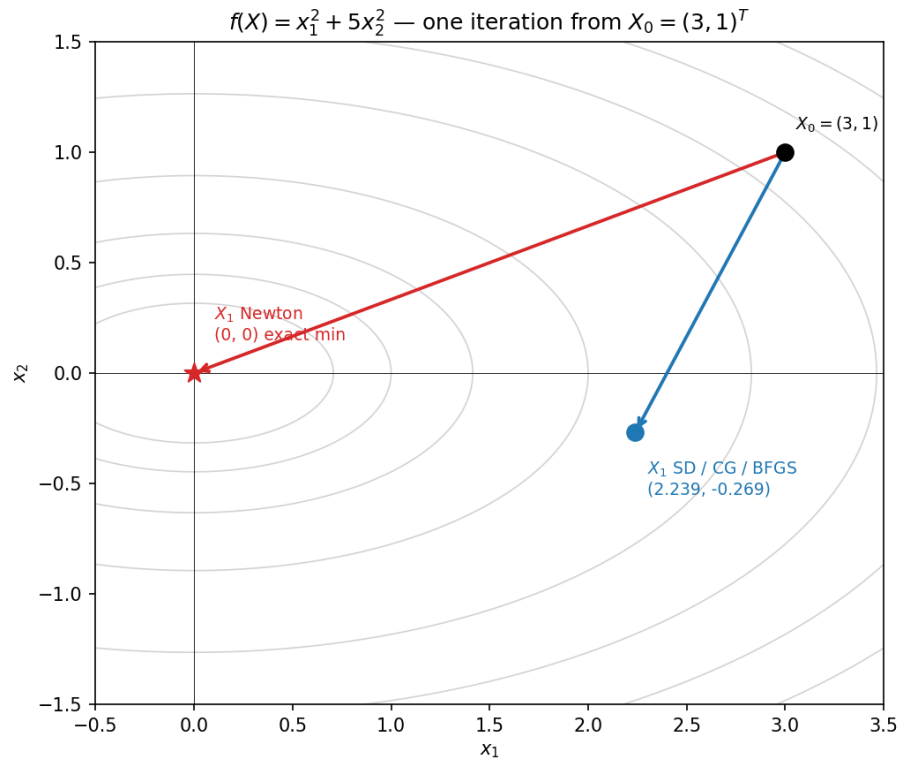
Result: $X_2 = \begin{bmatrix} 2.239 \\ -0.269 \end{bmatrix} + 0.3942 \begin{bmatrix} -5.681 \\ 0.682 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ **exact minimum in 2 iterations** (= n , same as CG)

Summary

Method	Direction p_0	α_0	X_1	Notes
Steepest descent	$-\nabla f(X_0)$	0.1269	$(2.239, -0.269)^T$	
Conjugate gradient	$-\nabla f(X_0) + \beta_0 p_{-1}$	0.1269	$(2.239, -0.269)^T$	$\beta_0 = 0$
Newton	$-H_f^{-1} \nabla f(X_0)$	1	$(0, 0)^T$	exact min
BFGS	$-\tilde{V}_0 \nabla f(X_0)$	0.1269	$(2.239, -0.269)^T$	$\tilde{V}_0 = I$

SD, CG, and BFGS all agree on iteration 1 because their direction formulas all reduce to $-\nabla f(X_0)$ at $k = 0$. Newton uses H_f^{-1} which accounts for the different curvature along x_1 and x_2 , allowing it to reach the minimum in one step.

Visualisation — Iterates on Contour Plot



Elliptical contours reflect the factor of 5 on x_2^2 . The condition number of H_f is $10/2 = 5$, causing the steepest descent step to overshoot across the valley. Newton corrects for this curvature exactly.